



Analyzing Pedestrian Accessibility to Parks and Green Spaces in Bronx Borough, NY

Course: AGOG 522 – GIS for Social Science
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IMPORTANCE OF GREEN SPACES

Public parks are essential for recreation, health, and community well-being. serve as vital public health infrastructure that provides physical, mental, and social benefits to millions of residents. Urban green spaces reduce cardiovascular disease, lower mortality rates, alleviate psychological stress, and promote physical activity among urban populations. Research shows neighborhoods with quality park access have higher physical activity rates, better mental health outcomes, and stronger community cohesion.

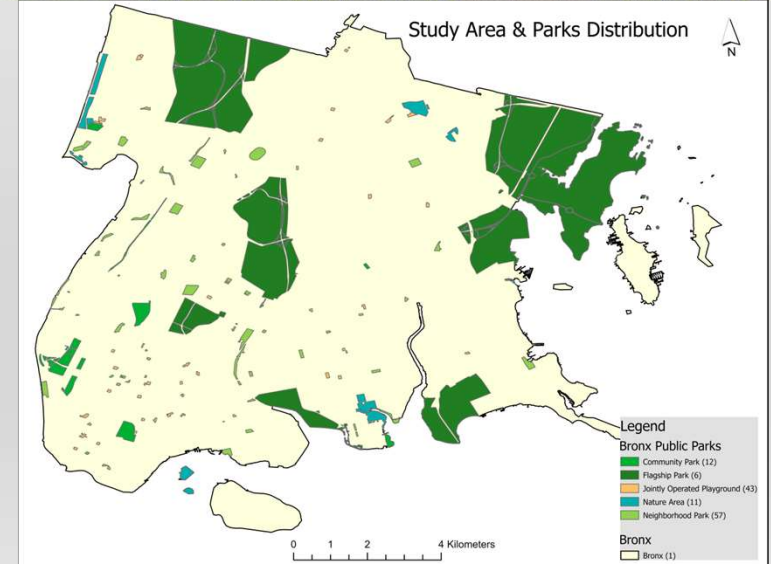
- Health: Reduce obesity, mental health issues, cardiovascular disease; provide recreation space
- Environmental: Improve air quality, reduce urban heat island effect, manage stormwater
- Social: Foster community cohesion, equitable access to nature, intergenerational spaces
- Economic: Increase property values, reduce healthcare costs, attract investment

Background

Studies show that living in greener environments reduces all-cause mortality, cardiovascular morbidity and symptoms of depression and anxiety while enhancing overall well-being and immune function

STUDY AREA

The Bronx is the northernmost borough of New York City, coextensive with Bronx County, and the only one primarily on the U.S. mainland, spanning 42 square miles with a population of about 1.47 million as of the 2020 census



RESEARCH QUESTION

Main Question: To what extent do residents of the Bronx have walkable access to public parks and green spaces, and which areas lack adequate accessibility?

METHODOLOGY

- Park Facilities: NYC Parks GIS database (2024), including park boundaries and locations
- Census Boundaries: 2024 TIGER/Line census tract boundaries (Bronx County FIPS 005 only)
- Demographic Data: American Community Survey (ACS) 5-year estimates, 2019–2023 (race/ethnicity, children <18, population)
- All data projected to: NAD83 / UTM Zone 18N (EPSG:26918)

Buffer Analysis: 400m (5min walk) and 800m (10 min walk) buffers around all public parks representing straight-line walking distances

Spatial Intersection: Overlay park buffers with census tract boundaries to identify:

- Accessible tracts: Census tracts with any overlap to park buffer
- Underserved tracts: Census tracts with zero overlap (no walkable park access)

Accessibility: Calculate total population living in accessible vs. underserved areas

RESULTS - ACCESS BUFFERS (400 M & 800 M)

Summary of Findings

Buffer Analysis Findings:

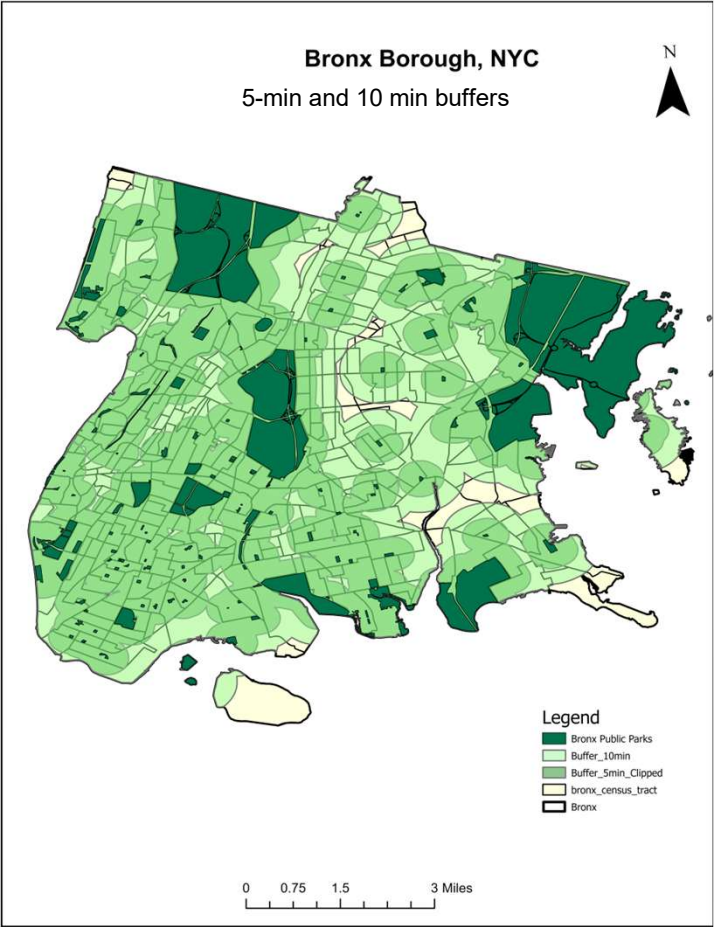
- The study evaluates pedestrian access to Bronx parks using two straight-line buffers: 400 m (≈5-minute walk) and 800 m (≈10-minute walk).
- The 400 m buffer represents immediate neighborhood access, while the 800 m buffer reflects a more generous planning standard for walkable parks.
- Census tracts are “underserved” based on whether they intersect these buffers, with the threshold used to define underserved areas

Method 400m buffer	Metric	Value / % of Bronx
Euclidean Buffer	Tracts with any park access	275 of 292 (94.2%)
Euclidean Buffer	Underserved tracts (no overlap)	17 of 292 (5.8%)
Euclidean Buffer	Population in underserved tracts	~40,000 residents (~3–4%)

400 m buffer approximates “immediate neighborhood” access

800 m buffer approximates extended, 10-minute walking access

Underserved definition based on 400 m

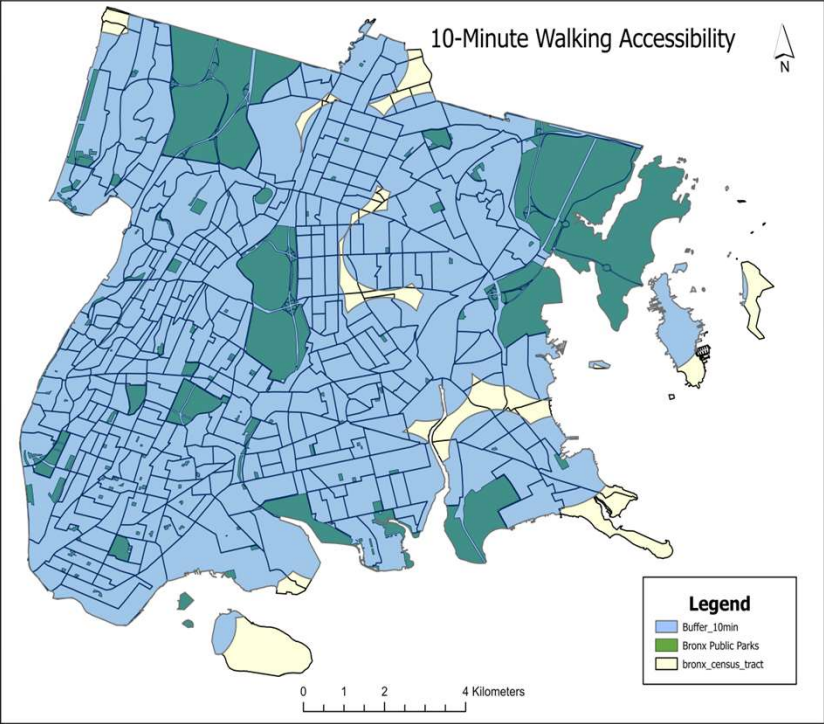


RESULTS: 10-MINUTE WALKING ACCESSIBILITY

Metric	400 m (5 min)	800 m (10 min, qualitative)
Visual coverage on map	Dense but with gaps	Nearly continuous coverage
Underserved tracts (no 400 m access)	17 tracts	Many gain nominal 800 m access but still far from parks
Interpretation	Immediate everyday access	Extended but more demanding walk

- 10-minute standard often used in planning
- In Bronx, 800 m enlarges the served area but does not fully resolve equity concerns

Using the map, show how the blue 10-minute service area extends across almost the entire borough, especially along the central spine and northern neighborhoods. Point out that some of the 17 underserved tracts from the 400 m analysis now sit within the 800 m buffer, meaning residents have at least one park within a longer walk. But emphasize two caveats. First, Euclidean buffers assume straight-line travel and ignore real-world barriers like highways, rail corridors, and limited park entrances, so even 800 m straight-line may underestimate how far people actually have to walk. Second, a 10-minute walk may be acceptable for some adults but less realistic for young children, seniors, or people with mobility limitations, especially in winter or on unsafe streets

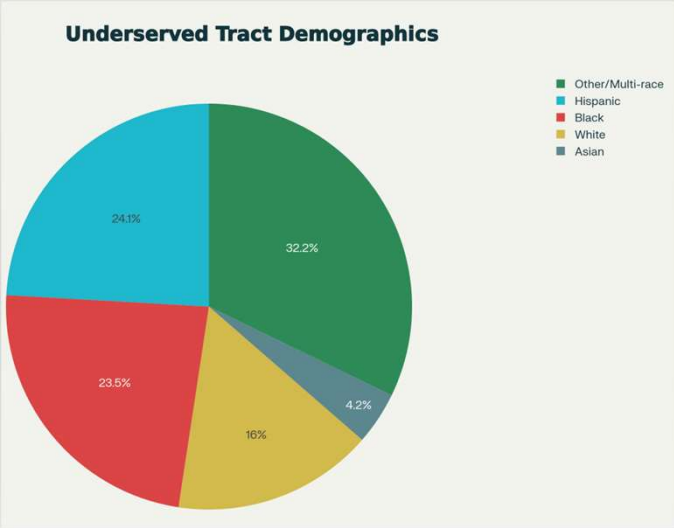
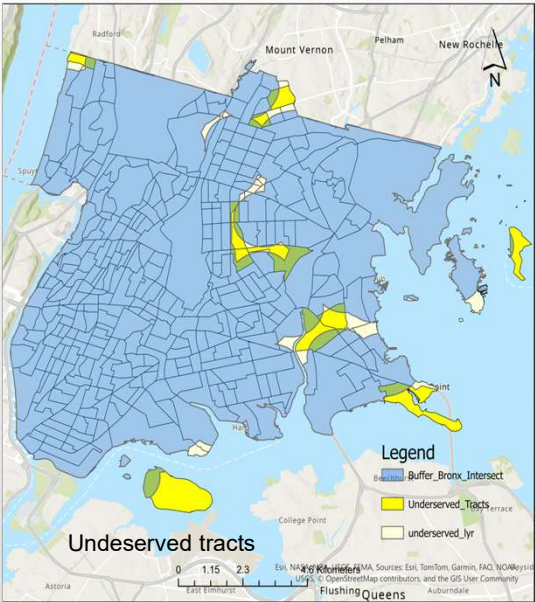


RESULTS - Results: Underserved Tracts & Who Lives There

Metric	Value
Number of underserved tracts (400 m)	17
Total population in these tracts	~40,000 residents
Share of Bronx tracts	5.8%
Share of Bronx population	~3–4%

Group	Share in underserved tracts
Hispanic	24.1%
Black	23.5%
White	16.0%
Asian	4.2%
Other/Multi	32.2%

The demographic table indicates who those ~40,000 residents are. Roughly 24.1% are Hispanic and 23.5% are Black, so almost half of the population in underserved tracts belongs to these two minority groups. About 16% are White and 4.2% are Asian, while a large share 32.2% falls into the Other/Multi category, suggesting substantial diversity and mixed racial/ethnic identities.



DISCUSSION

Urban Form and Land Use

Urban form in the Bronx features a mix of large regional parks like Pelham Bay Park and smaller neighborhood parks, creating uneven access patterns. Large parks serve broader recreation but often require longer travel, while small parks provide local convenience yet may lack amenities. This duality shapes accessibility, with denser urban land use limiting park integration in residential areas

Infrastructure Barriers

Highways such as the Cross Bronx Expressway and Sheridan Expressway, along with rail corridors and waterfront edges, physically fragment neighborhoods and block pedestrian paths to parks. These barriers, remnants of mid-20th-century planning, isolate communities from green spaces like the Bronx River waterfront.

Street Network and Entrances

The Bronx's irregular street grids, combined with limited park entry points, hinder walkable access, forcing detours around barriers. Natural features like the Bronx River and infrastructure split crosstown connections, increasing travel times. Poorly placed entrances compound this, particularly near highways and rail

Social and Economic Context

Underserved tracts in the Bronx concentrate vulnerable populations, including low-income families, seniors, and children, who face higher barriers to parks amid health disparities. Areas like the South and West Bronx show inequities, with longer travel times to green spaces correlating to limited physical activity. These patterns highlight environmental justice issues, as parks reduce obesity risks but remain inaccessible

CONCLUSIONS

Key Findings:

- 94.2% of Bronx tracts have some park access within 400 m, yet 17 tracts remain completely outside this 5-minute catchment.
- These underserved tracts contain ~40,000 residents, including substantial shares of minority populations and children.
- Extending to 800 m suggests almost universal nominal access, but straight-line buffers likely overestimate true pedestrian accessibility in dense, barrier-filled environments.
- Bronx has a strong park network overall, but a measurable minority of residents still live in neighborhoods without a nearby, easily walkable park.
- Using 400 m vs. 800 m thresholds, and relying on Euclidean rather than network distance, can shift tracts in or out of the “served” category, changing which communities appear as priorities for investment.

RECOMMENDATIONS

Target Underserved Tracts:

- Prioritize park creation or expansion in the 17 400 m-underserved tracts; explore schoolyards, vacant lots, or small pocket parks as opportunities.

Improve Connectivity and Entrances:

- Enhance pedestrian connections (crosswalks, sidewalks, lighting) between underserved tracts and nearby existing parks.
- Add or redesign park entrances where buffers reach neighborhoods but access points are limited.

Integrate Accessibility into Equity Planning:

- Use the 400 m analysis as an annual monitoring indicator for the Bronx.
- Combine accessibility metrics with demographic data to identify communities facing cumulative environmental burdens

THANKYOU